

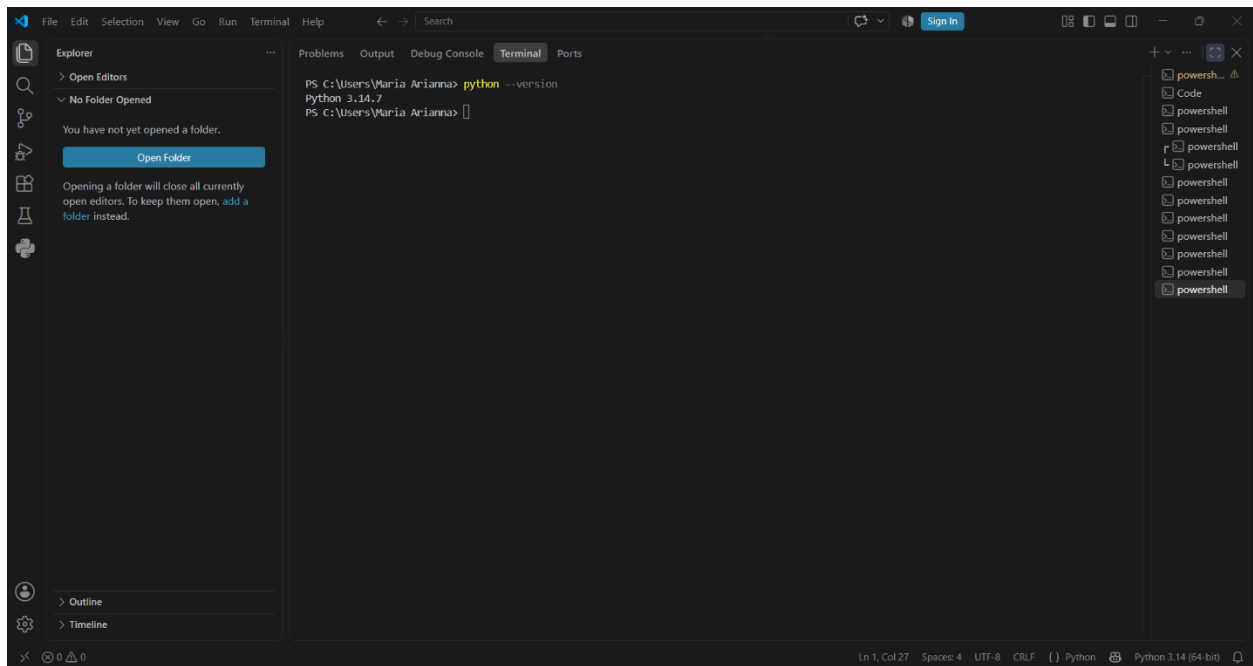


Figure 1. The terminal confirms the installed Python version used for the numerical methods laboratory.

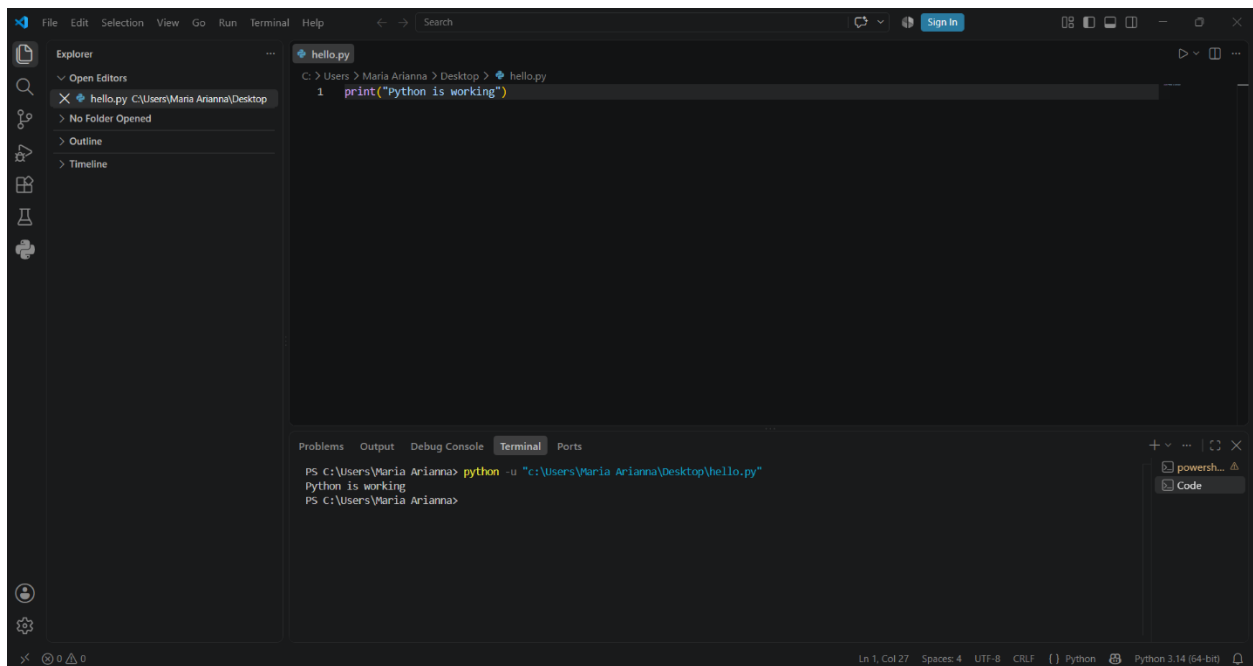


Figure 2. The hello.py program is displayed in VS Code with its successful output shown in the terminal.

```
PS C:\Users\Maria Arianna> python -m pip install --upgrade pip
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pip in c:\program files\python314\lib\site-packages (26.2.1)
PS C:\Users\Maria Arianna> python -m pip install numpy scipy matplotlib
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: numpy in .AppData\Roaming\Python\Python314\site-packages (2.5.2)
Requirement already satisfied: scipy in .AppData\Roaming\Python\Python314\site-packages (1.18.0)
Requirement already satisfied: matplotlib in .AppData\Roaming\Python\Python314\site-packages (3.11.1)
Requirement already satisfied: contourpy>=1.0.1 in .AppData\Roaming\Python\Python314\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in .AppData\Roaming\Python\Python314\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in .AppData\Roaming\Python\Python314\site-packages (from matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in .AppData\Roaming\Python\Python314\site-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in .AppData\Roaming\Python\Python314\site-packages (from matplotlib) (26.3)
Requirement already satisfied: pillow>=9 in .AppData\Roaming\Python\Python314\site-packages (from matplotlib) (12.3.0)
Requirement already satisfied: pyparsing>=3 in .AppData\Roaming\Python\Python314\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in .AppData\Roaming\Python\Python314\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in .AppData\Roaming\Python\Python314\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
PS C:\Users\Maria Arianna>
```

Figure 3. The terminal confirms that NumPy, SciPy, and Matplotlib are installed and available for use.

```
1 ***
2 Lab Assignment 0 - platform verification
3 Name : <Maria Arianna Chriselle I. Samson>
4 Student number : <TUPM-23-0026>
5 ***
6 import os
7 import sys
8
9 import numpy as np
10 import scipy
11 from scipy import integrate
12 import matplotlib
13 import matplotlib.pyplot as plt
14
15 # ---- 1. report the versions ----
16 print("Python :", sys.version.split()[0])
17 print("Numpy :", np.__version__)
18 print("Scipy :", scipy.__version__)
19 print("Matplotlib :", matplotlib.__version__)
20
21 # ---- 2. a Numpy calculation ----
22 # 200 points from 0 to 2*pi, then the sine of every one of them at once.

PS C:\Users\Maria Arianna\Desktop> python lab00_check_Samson.py

Python : 3.14.7
Numpy : 2.5.2
Scipy : 1.18.0
Matplotlib : 3.11.1

mean of sin(x) over one period : 0.0
integral of sin(x) from 0 to pi : 2.0 (exact value 2)

figure written to figures/lab00_check.png
PLATFORM READY.
PS C:\Users\Maria Arianna\Desktop>
```

Figure 4. The verification program successfully tests Python, NumPy, SciPy, and Matplotlib and ends with “PLATFORM READY.”

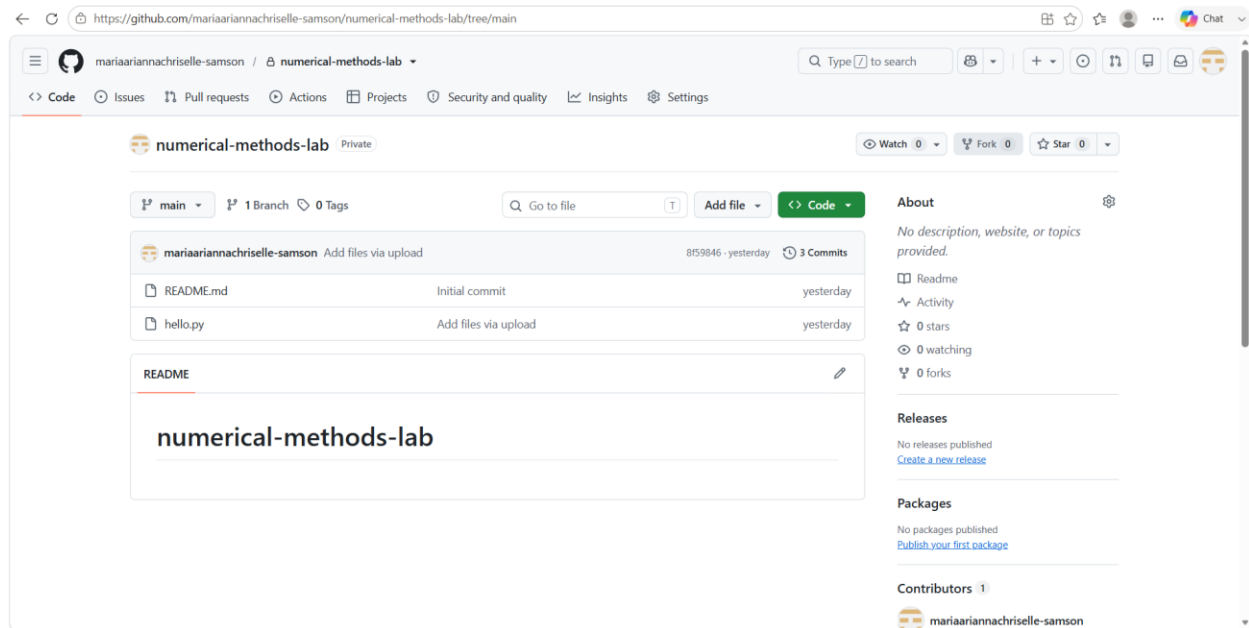


Figure 5. The private GitHub repository numerical-methods-lab contains the uploaded hello.py file.

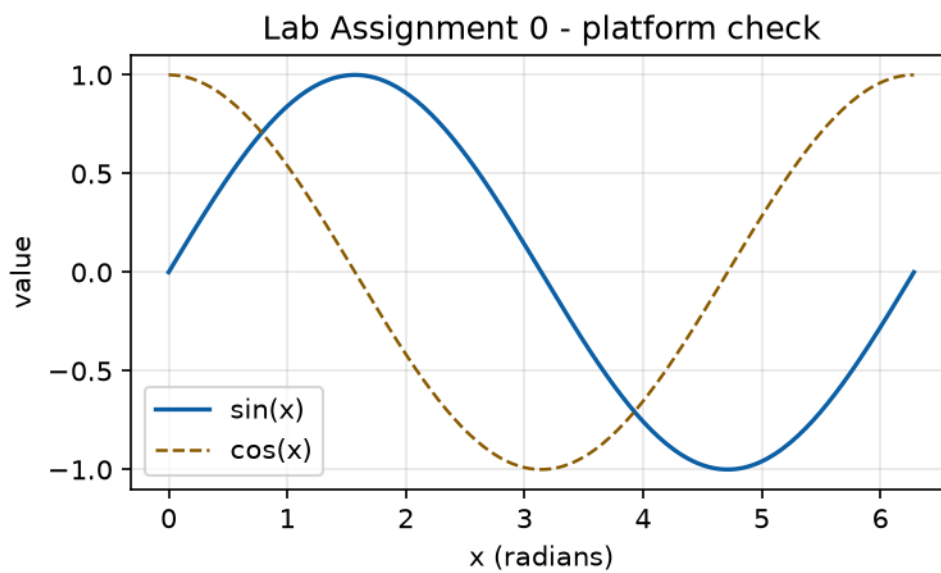


Figure 6. Generated sine and cosine graph from the verification program.